

# CBSE Class 12 Chemistry — Haloalkanes and Haloarenes

## 50 Most-Probable PYQs & Board-Pattern Practice Questions

### 1-Mark Questions (MCQ / Assertion-Reason / VSA)

- Q1.** Which isomer of pentane ( $C_5H_{12}$ ) will give only a single monochloride upon photochemical chlorination? (a) n-Pentane (b) Isopentane (c) Neopentane (d) None of the above [PYQ Term-1 2021] [Confidence: Very High]
- Q2.** The major product obtained on the reaction of 3-Phenylpropene with  $HBr$  in the presence of an organic peroxide is: (a) 3-phenyl-1-bromopropane (b) 1-phenyl-3-bromopropane (c) 1-phenyl-2-bromopropane (d) 3-phenyl-2-bromopropane [PYQ SQP Term-1 2021] [Confidence: High]
- Q3.** The reaction of toluene with  $Cl_2$  in the presence of  $FeCl_3$  gives 'X', while the reaction of toluene with  $Cl_2$  in the presence of sunlight gives 'Y'. Identify 'X' and 'Y'. (a) X = benzyl chloride, Y = o- and p-chlorotoluene (b) X = m-chlorotoluene, Y = p-chlorotoluene (c) X = o- and p-chlorotoluene, Y = trichloromethylbenzene/benzyl chloride (d) X = benzyl chloride, Y = m-chlorotoluene [PYQ SQP Term-1 2021] [Confidence: Very High]
- Q4.** Which specific reagents are required for the one-step conversion of chlorobenzene to toluene (Wurtz-Fittig reaction)? (a)  $CH_3Cl/AlCl_3$  (b)  $CH_3Cl, Na, Dry ether$  (c)  $CH_3Cl/Fe dark$  (d)  $NaNO_2/HCl/0 - 5^\circ C$  [PYQ SQP Term-1 2021] [Confidence: Very High]
- Q5.** Which one of the following compounds is most reactive towards an  $S_N1$  reaction? (a)  $CH_2 = CHCH_2Br$  (b)  $C_6H_5CH_2Br$  (c)  $C_6H_5CH(C_6H_5)Br$  (d)  $C_6H_5CH(CH_3)Br$  [PYQ SQP 2022] [Confidence: Very High]
- Q6.** Which of the following isomeric dichlorobenzenes has the highest melting point? (a) 1,2-Dichlorobenzene (b) 1,3-Dichlorobenzene (c) 1,4-Dichlorobenzene (d) All isomers have the same melting point [PYQ SQP Term-1 2021] [Confidence: Very High]
- Q7.** Enantiomers (optical isomers) differ from each other only in their: (a) Boiling point (b) Rotation of plane-polarised light (c) Melting point (d) Solubility [PYQ Term-1 2021] [Confidence: High]
- Q8.** Which of the following haloalkanes has the highest boiling point? (a)  $C_2H_5F$  (b)  $C_2H_5Cl$  (c)  $C_2H_5Br$  (d)  $C_2H_5I$  [PYQ Term-1 2021] [Confidence: Very High]
- Q9.** Which of the following alkyl halides will undergo an  $S_N1$  reaction most readily? (a)  $(CH_3)_3C - Cl$  (b)  $(CH_3)_3C - Br$  (c)  $(CH_3)_3C - F$  (d)  $(CH_3)_3C - I$  [PYQ SQP Term-1 2021] [Confidence: Very High]

**Q10.** The process of racemisation strictly occurs in which of the following reactions of an optically active alkyl halide? (a)  $S_N2$  reaction (b)  $S_N1$  reaction (c) Neither  $S_N1$  nor  $S_N2$  reaction (d) Both  $S_N1$  and  $S_N2$  reactions [PYQ 2020] [Confidence: Very High]

**Q11.** The conversion of an alkyl halide into an alcohol by heating it with aqueous  $NaOH$  is chemically classified as: (a) A dehydrohalogenation reaction (b) A nucleophilic substitution reaction (c) An addition reaction (d) A dehydration reaction [PYQ 2020] [Confidence: Very High]

**Q12.** Assertion (A): Chlorobenzene is highly resistant to electrophilic substitution reactions and does not undergo them at all. Reason (R): The  $C - Cl$  bond in chlorobenzene acquires a partial double bond character due to resonance. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2023] [Confidence: High]

**Q13.** Assertion (A): Alkyl halides are completely insoluble in water despite being polar molecules. Reason (R): Alkyl halides have their halogen atom attached to an  $sp^3$  hybridised carbon. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ SQP Term-1 2021] [Confidence: Very High]

**Q14.** Out of chloromethane ( $CH_3Cl$ ) and fluoromethane ( $CH_3F$ ), which one possesses a higher dipole moment and why? [PYQ SQP 2022] [Confidence: High]

**Q15.** Write the structural formula of 3-Bromo-2-methylprop-1-ene. [PYQ 2017] [Confidence: Very High]

**Q16.** Identify the chiral molecule out of the following pair: 1-chlorobutane and 2-chlorobutane. [PYQ 2014] [Confidence: Very High]

**Q17.** Write the exact IUPAC name of the compound:  $CH_2 = CHCH_2Br$ . [PYQ 2014] [Confidence: Very High]

**Q18.** What are ambident nucleophiles? Give one common example. [PYQ 2014] [Confidence: High]

**Q19.** Which of the following compounds is an example of an allylic halide? (a)  $CH_2 = CH - Cl$  (b)  $C_6H_5 - CH_2 - Cl$  (c)  $CH_2 = CH - CH_2 - Cl$  (d)  $C_6H_5 - Cl$  [PYQ 2017] [Confidence: Very High]

**Q20.** Complete stereochemical inversion of configuration is the hallmark of which nucleophilic substitution mechanism? [Practice] [Confidence: Very High]

## 2-Mark Questions (Short Answer)

**Q21.** Why is the thionyl chloride ( $SOCl_2$ ) method preferred over other reagents for preparing alkyl chlorides from alcohols? [PYQ 2019] [Confidence: Very High]

**Q22.** Explain why electrophilic substitution reactions in haloarenes occur slowly and require more drastic conditions as compared to those in plain benzene. [PYQ SQP 2023] [Confidence: Very High]

**Q23.** Arrange the following compounds in the increasing order of their reactivity towards an  $S_N2$  reaction mechanism, and state the reason: 2-Bromopentane, 1-Bromopentane, 2-Bromo-2-methylbutane [PYQ 2023] [Confidence: Very High]

**Q24.** Account for the following: p-dichlorobenzene possesses a noticeably higher melting point than its corresponding ortho- and meta-isomers. [PYQ 2023] [Confidence: Very High]

**Q25.** Name the suitable alcohol and the best reagent from which 2-chloro-2-methylpropane can be quickly prepared at room temperature. [PYQ SQP 2022] [Confidence: Medium]

**Q26.** Out of chlorobenzene and benzyl chloride, which one gets easily hydrolysed by aqueous  $NaOH$  to form an alcohol, and why? [PYQ 2018] [Confidence: Very High]

**Q27.** Explain why alkyl halides, even though they are strongly polar in nature, remain essentially immiscible (insoluble) with water. [PYQ 2017] [Confidence: Very High]

**Q28.** Explain structurally why the dipole moment of chlorobenzene is notably lower than that of cyclohexyl chloride. [PYQ 2016] [Confidence: Very High]

**Q29.** Give detailed reasons for the following observations: (a) The  $C - Cl$  bond length in chlorobenzene is shorter than the  $C - Cl$  bond length in  $CH_3Cl$ . (b)  $S_N1$  reactions are characteristically accompanied by racemisation in optically active alkyl halides. [PYQ 2016] [Confidence: Very High]

**Q30.** How will you seamlessly convert the following? (a) Chlorobenzene to biphenyl (b) 2-bromobutane to but-2-ene [PYQ 2016] [Confidence: High]

**Q31.** Give a strong reason why n-Butyl bromide has a higher boiling point than its branched isomer, t-butyl bromide. [PYQ 2015] [Confidence: High]

**Q32.** Which compound would undergo an  $S_N2$  reaction faster in the following pair and why?  $CH_3 - CH_2 - Br$  OR  $CH_3 - CH_2 - I$  [PYQ 2015] [Confidence: Very High]

### 3-Mark Questions (Short Answer)

**Q33.** Write the balanced chemical equations for the reactions that take place when: (a) Methyl chloride is treated with  $AgNO_2$ . (b) Bromobenzene is treated with  $CH_3Cl$  in the presence of anhydrous  $AlCl_3$ . (c) Ethyl chloride is heated with aqueous  $KOH$ . [PYQ 2014] [Confidence: Very High]

**Q34.** Provide strong chemical reasoning for the following: (a) Haloalkanes react with  $KCN$  to form alkyl cyanides as the main product, while  $AgCN$  strictly forms isocyanides. (b) Grignard reagents must always be prepared under strictly anhydrous conditions. (c) Aryl halides are extremely less reactive towards nucleophilic substitution compared to alkyl halides. [Practice / PYQ themes] [Confidence: Very High]

**Q35.** How are the following organic conversions carried out in not more than two steps? (a) Propene to 1-iodopropane (b) Benzyl chloride to benzyl alcohol (c) Toluene to benzyl alcohol [PYQ 2016 / Practice] [Confidence: High]

**Q36.** Aryl chlorides and bromides can be easily prepared by the electrophilic substitution of arenes with chlorine and bromine. However, the preparation of aryl iodides requires the

mandatory presence of an oxidising agent (like  $HIO_4$  or  $HNO_3$ ). Explain why. [Practice]  
[Confidence: Medium]

**Q37.** Classify the following compounds systematically as primary, secondary, and tertiary halides: (i) 1-Bromobut-2-ene (ii) 4-Bromopent-2-ene (iii) 2-Bromo-2-methylpropane [Practice]  
[Confidence: High]

**Q38.** Write the exact structure of the major organic product formed in each of the following reactions: (a)  $CH_3CH = C(CH_3)_2 + HBr \rightarrow$  (b)  $CH_3CH_2CH_2OH + SOCl_2 \rightarrow$  (c)  $CH_3CH_2CH_2Cl + NaI \xrightarrow{\text{dry acetone}} \rightarrow$  [PYQ 2015] [Confidence: Very High]

**Q39.** Compound 'A' with the molecular formula  $C_4H_9Br$  is treated with aqueous  $KOH$  solution. The rate of this reaction strictly depends upon the concentration of compound 'A' only. When another optically active isomer 'B' of this same compound was treated with aqueous  $KOH$ , the rate of reaction was found to be dependent on the concentration of both the compound and  $KOH$ . Identify the structures of 'A' and 'B', and specify their respective reaction mechanisms. [Practice] [Confidence: High]

**Q40.** Predict the major organic product formed when  $HCl$  is added to isobutylene (2-methylprop-1-ene). Explain the electrophilic addition mechanism involved, identifying the stable intermediate. [Practice] [Confidence: High]

**Q41.** Give the exact IUPAC names for the following complex organic halides: (a)  $(CH_3)_3CCH_2Br$  (b)  $CH_3CH(Cl)CH(Br)CH_3$  (c)  $p - ClC_6H_4CH_2CH(CH_3)_2$  [Practice] [Confidence: Medium]

**Q42.** Write the specific chemical equations (with appropriate reagents and conditions) to illustrate the following classic named reactions: (a) Finkelstein reaction (b) Swarts reaction (c) Wurtz-Fittig reaction [Practice] [Confidence: Very High]

### 5-Mark Questions (Long Answer / Case-Study)

**Q43.** (a) Write the detailed step-by-step mechanism of an  $S_N1$  reaction using the hydrolysis of tert-butyl bromide as a suitable example. (2 Marks) (b) Out of 1-bromobutane and 2-bromobutane, state which one is chiral and define chirality. (2 Marks) (c) State one major difference between  $S_N1$  and  $S_N2$  reactions explicitly regarding their stereochemical outcomes. (1 Mark) [PYQ 2020 / Practice] [Confidence: Very High]

**Q44.** (a) Some alkyl halides undergo nucleophilic substitution whereas some undergo elimination upon treatment with bases. Discuss the structural features of alkyl halides and the nature of the base (aqueous vs alcoholic) responsible for determining this competing difference. (3 Marks) (b) Predict the exact order of reactivity of the following compounds towards an  $S_N2$  nucleophilic substitution reaction, giving a brief reason:  $CH_3Br$ ,  $(CH_3)_2CHBr$ ,  $(CH_3)_3CBr$ ,  $CH_3CH_2Br$ . (2 Marks) [Practice] [Confidence: High]

**Q45. Case Study: Nucleophilic Substitution in Haloalkanes** Alkyl halides undergo nucleophilic substitution reactions rapidly. The mechanism can follow either an  $S_N1$  pathway (unimolecular) or an  $S_N2$  pathway (bimolecular) heavily depending on the structure of the alkyl halide, the polarity of the solvent, and the strength of the incoming nucleophile. (i) What

is the correct order of reactivity of primary, secondary, and tertiary alkyl halides strictly for an  $S_N1$  reaction? (1 Mark) (ii) Name the general class of solvents (e.g., polar protic or polar aprotic) that fundamentally favours  $S_N2$  reactions. (1 Mark) (iii) Identify the mechanism followed when optically active 2-bromobutane reacts with aqueous  $KOH$  to form a racemic mixture. Explain clearly why racemisation occurs in this specific case. (3 Marks) [Practice] [Confidence: Very High]

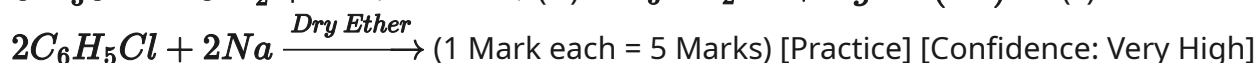
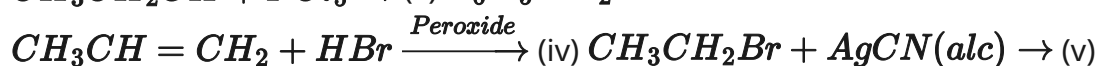
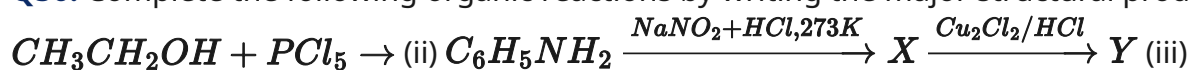
**Q46.** (a) Explain chemically why the  $S_N1$  mechanism is heavily preferred for the hydrolysis of tert-butyl bromide, while the  $S_N2$  mechanism is strictly preferred for the hydrolysis of methyl bromide. (2 Marks) (b) Write complete chemical equations to bring about the following conversions: (i) Aniline to Chlorobenzene (ii) Ethanol to But-1-yne (iii) 1-Chlorobutane to n-octane (3 Marks) [PYQ Themes] [Confidence: High]

**Q47. Case Study: Elimination Reactions in Haloalkanes** When a haloalkane containing  $\beta$ -hydrogen atoms is heated with an alcoholic solution of potassium hydroxide, there is an aggressive elimination of a hydrogen halide molecule, resulting in the formation of an alkene. This is known as  $\beta$ -elimination. (i) Name the empirical rule specifically used to predict the major product in the dehydrohalogenation of unsymmetrical haloalkanes. (1 Mark) (ii) What is the IUPAC name of the major alkene product formed when 2-Bromopentane is heavily treated with alcoholic  $KOH$ ? (1 Mark) (iii) Write the complete balanced equation for the reaction in part (ii). Explain using the rule from part (i) why this specific alkene dominates as the major product. (3 Marks) [Practice] [Confidence: Very High]

**Q48.** Account for the following observations thoroughly: (i) Haloalkanes easily dissolve in organic solvents but are only very slightly soluble in water. (1 Mark) (ii) The presence of electron-withdrawing groups (like  $-NO_2$ ) at ortho and para positions heavily increases the reactivity of haloarenes towards nucleophilic substitution. (1 Mark) (iii) Grignard reagents ( $RMgX$ ) are highly reactive and react rapidly with any source of protons. (1 Mark) (iv) Optically active alkyl halides undergoing solely  $S_N2$  reactions show a 100% complete inversion of configuration. (1 Mark) (v) Unlike haloalkanes, haloarenes do not undergo Friedel-Crafts alkylation easily without a catalyst, and the halogen atom acts as a deactivating but ortho-para directing group. (1 Mark) [Practice] [Confidence: High]

**Q49.** (a) Write the specific chemical reactions involved in: (i) Sandmeyer reaction (starting from aniline) (ii) Friedel-Crafts alkylation of chlorobenzene (with  $CH_3Cl$ ) (2 Marks) (b) An unknown organic compound 'A' having molecular formula  $C_4H_9Cl$  on reaction with  $Na$  metal in dry ether yields a symmetrical hydrocarbon 'B' ( $C_8H_{18}$ ). Hydrocarbon 'B' on photochemical monochlorination gives only one single unique chloro derivative. Identify the exact structures and IUPAC names of A and B. (3 Marks) [PYQ / Practice] [Confidence: Very High]

**Q50.** Complete the following organic reactions by writing the major structural products: (i)





## Answer Key

**Q1.** (c) Neopentane (All 12 hydrogens are chemically equivalent). **Q2.** (b) 1-phenyl-3-bromopropane (Anti-Markovnikov addition). **Q3.** (c) X = o- and p-chlorotoluene (Electrophilic substitution in ring), Y = benzyl chloride (Free radical substitution in side chain). **Q4.** (b)  $\text{CH}_3\text{Cl}$ ,  $\text{Na}$ , *Dry ether* (Wurtz-Fittig reaction). **Q5.** (c)  $\text{C}_6\text{H}_5\text{CH}(\text{C}_6\text{H}_5)\text{Br}$  (Yields highly resonance-stabilized diphenylmethyl carbocation). **Q6.** (c) 1,4-Dichlorobenzene (Symmetrical structure fits perfectly into the crystal lattice). **Q7.** (b) Rotation of plane-polarised light. **Q8.** (d)  $\text{C}_2\text{H}_5\text{I}$  (Highest molecular mass and largest van der Waals forces due to iodine). **Q9.** (d)  $(\text{CH}_3)_3\text{C} - \text{I}$  (Iodide is the best leaving group; forms stable tertiary carbocation). **Q10.** (b)  $\text{S}_{\text{N}}1$  reaction. **Q11.** (b) A nucleophilic substitution reaction. **Q12.** (d) A is false but R is true. (It undergoes electrophilic substitution slowly, it is not fully resistant). **Q13.** (b) Both A and R are true but R is not the correct explanation of A. (Insolubility is due to inability to form H-bonds with water). **Q14.** Chloromethane ( $\text{CH}_3\text{Cl}$ ). The  $\text{C} - \text{Cl}$  bond length is significantly larger than  $\text{C} - \text{F}$ , which compensates for fluorine's higher electronegativity ( $\text{Dipole} = \text{charge} \times \text{distance}$ ). **Q15.**  $\text{CH}_2 = \text{C}(\text{CH}_3) - \text{CH}_2\text{Br}$ . **Q16.** 2-chlorobutane (Its 2nd carbon is attached to four different groups: -H, -Cl, -CH<sub>3</sub>, -C<sub>2</sub>H<sub>5</sub>). **Q17.** 3-Bromoprop-1-ene. **Q18.** Nucleophiles that possess two different nucleophilic donor centres (e.g.,  $\text{CN}^-$  can attack via C or N). **Q19.** (c)  $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{Cl}$  (Halogen is attached to an  $\text{sp}^3$  carbon adjacent to a  $\text{C} = \text{C}$  double bond). **Q20.**  $\text{S}_{\text{N}}2$  mechanism.

**Q21.** The by-products of the reaction,  $\text{SO}_2$  and  $\text{HCl}$ , are both escapable gases, leaving behind pure alkyl chloride without needing complex separation. **Q22.** The halogen atom pulls electron density away from the benzene ring via the -I effect, deactivating the ring and making it less susceptible to electrophilic attack. **Q23.** 2-Bromo-2-methylbutane < 2-Bromopentane < 1-Bromopentane.  $\text{S}_{\text{N}}2$  reactivity strictly decreases as steric hindrance around the  $\alpha$ -carbon increases ( $3^\circ < 2^\circ < 1^\circ$ ). **Q24.** p-Dichlorobenzene is highly symmetrical, allowing its molecules to pack closely and efficiently in the solid crystal lattice, requiring more thermal energy to break. **Q25.** 2-Methylpropan-2-ol (tert-butyl alcohol). Reagent: Concentrated  $\text{HCl}$  and anhydrous  $\text{ZnCl}_2$  (Lucas Reagent). **Q26.** Benzyl chloride. It undergoes  $\text{S}_{\text{N}}1$  hydrolysis easily because the resulting benzyl carbocation is highly stabilized by resonance. **Q27.** To dissolve in water, they must break strong hydrogen bonds between water molecules, but the new dipole-dipole interactions between haloalkane and water are too weak to compensate for the energy required. **Q28.** In chlorobenzene, the  $\text{C} - \text{Cl}$  bond is attached to an  $\text{sp}^2$  hybridized carbon (more electronegative), while in cyclohexyl chloride it's attached to an  $\text{sp}^3$  carbon. The  $\text{sp}^2$  carbon reduces the overall polarity and bond length. **Q29.** (a) Resonance delocalization of chlorine's lone pairs into the benzene ring imparts partial double bond character to the  $\text{C} - \text{Cl}$  bond. (b) The intermediate carbocation is planar, allowing nucleophilic attack from both front and back faces with equal probability. **Q30.** (a) Fittig reaction: Treat with  $\text{Na}$  metal in dry ether. (b) Treat with Alcoholic  $\text{KOH}$  and heat (Dehydrohalogenation). **Q31.** n-Butyl bromide has a straight-chain structure giving it a larger surface area and stronger van der Waals forces compared to the spherical, compact shape of t-butyl bromide. **Q32.**  $\text{CH}_3\text{CH}_2\text{I}$  reacts faster. The iodide ion ( $\text{I}^-$ ) is much larger and a

significantly better leaving group than the bromide ion ( $\text{Br}^-$ ) due to its lower bond dissociation enthalpy.

**Q33.** (a)  $\text{CH}_3\text{Cl} + \text{AgNO}_2 \rightarrow \text{CH}_3 - \text{NO}_2$  (Nitromethane) +  $\text{AgCl} \downarrow$  (b)

$\text{C}_6\text{H}_5\text{Br} + \text{CH}_3\text{Cl} \xrightarrow{\text{Anhy. AlCl}_3} o\text{-bromo-toluene} + p\text{-bromo-toluene}$  (c)

$\text{CH}_3\text{CH}_2\text{Cl} + \text{KOH(aq)} \xrightarrow{\Delta} \text{CH}_3\text{CH}_2\text{OH} + \text{KCl}$  **Q34.** (a) KCN is predominantly ionic, providing  $\text{CN}^-$  which attacks via the less electronegative Carbon to form stable C-C bonds. AgCN is covalent, forcing attack via the lone pair on Nitrogen, forming isocyanides. (b) They are highly basic and react instantly with water to form alkanes (

$\text{RMgX} + \text{H}_2\text{O} \rightarrow \text{RH} + \text{Mg(OH)X}$ ). (c) Partial double-bond character of C-X bond,  $sp^2$  hybridization of C, and instability of the phenyl cation. **Q35.** (a)

$\text{CH}_3 - \text{CH} = \text{CH}_2 + \text{HBr/Peroxide} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ . Then

$\text{CH}_3\text{CH}_2\text{CH}_2\text{Br} + \text{NaI} \xrightarrow{\text{acetone}} \text{CH}_3\text{CH}_2\text{CH}_2\text{I}$ . (b)

$\text{C}_6\text{H}_5\text{CH}_2\text{Cl} + \text{KOH(aq)} \xrightarrow{\Delta} \text{C}_6\text{H}_5\text{CH}_2\text{OH} + \text{KCl}$ . (c) Toluene +  $\text{Cl}_2/h\nu \rightarrow$  Benzyl chloride. Then treat with aq. KOH  $\rightarrow$  Benzyl alcohol. **Q36.** Iodination is chemically reversible.

The  $\text{HI}$  produced heavily pushes the reaction backward. An oxidizing agent ( $\text{HIO}_4$  or  $\text{HNO}_3$ ) oxidizes  $\text{HI}$  to  $\text{I}_2$ , preventing the backward reaction and driving it forward. **Q37.** (i)

Primary ( $1^\circ$ ) (ii) Secondary ( $2^\circ$ ) (iii) Tertiary ( $3^\circ$ ). **Q38.** (a)  $\text{CH}_3 - \text{CH}_2 - \text{C}(\text{Br})(\text{CH}_3)_2$

(Markovnikov addition) (b)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{SO}_2 \uparrow + \text{HCl} \uparrow$  (c)

$\text{CH}_3\text{CH}_2\text{CH}_2\text{I} + \text{NaCl} \downarrow$  **Q39.** A undergoes  $\text{S}_\text{N}1$  (rate depends on A), meaning A forms a stable carbocation. A is 2-Bromo-2-methylpropane (Tertiary). B undergoes  $\text{S}_\text{N}2$  (rate depends on B and KOH). B is chiral and  $\text{S}_\text{N}2$  favorable, so B is 2-Bromobutane (Secondary). **Q40.** Major product is 2-chloro-2-methylpropane. Mechanism: The  $\pi$ -bond attacks  $\text{H}^+$  to form the most stable  $3^\circ$  carbocation intermediate (tert-butyl cation), which is then rapidly attacked by the  $\text{Cl}^-$  nucleophile. **Q41.** (a) 1-Bromo-2,2-dimethylpropane. (b) 2-Bromo-3-chlorobutane. (c) 1-Chloro-4-(2-methylpropyl)benzene. **Q42.** (a) Finkelstein:

$\text{R} - \text{X} + \text{NaI} \xrightarrow{\text{Dry Acetone}} \text{R} - \text{I} + \text{NaX} \downarrow$  (b) Swarts:

$\text{R} - \text{Br} + \text{AgF} \rightarrow \text{R} - \text{F} + \text{AgBr} \downarrow$  (c) Wurtz-Fittig:

$\text{Ar} - \text{X} + \text{R} - \text{X} + 2\text{Na} \xrightarrow{\text{Dry Ether}} \text{Ar} - \text{R} + 2\text{NaX}$

**Q43.** (a) Step 1:  $(\text{CH}_3)_3\text{C} - \text{Br} \rightleftharpoons (\text{CH}_3)_3\text{C}^+ + \text{Br}^-$  (Slow, RDS). Step 2:

$(\text{CH}_3)_3\text{C}^+ + \text{OH}^- \rightarrow (\text{CH}_3)_3\text{C} - \text{OH}$  (Fast). (b) 2-bromobutane is chiral because its carbon-2 is bonded to 4 different groups (H, Br, CH<sub>3</sub>, C<sub>2</sub>H<sub>5</sub>). Chirality is the property of non-superimposability on a mirror image. (c)  $\text{S}_\text{N}1$  leads to racemisation;  $\text{S}_\text{N}2$  leads to complete stereochemical inversion. **Q44.** (a) Aqueous KOH contains heavily hydrated  $\text{OH}^-$  ions acting strictly as nucleophiles (Substitution). Alcoholic KOH yields alkoxide ions ( $\text{RO}^-$ ), which are strong bases that abstract  $\beta$ -hydrogens to form alkenes (Elimination). Highly branched tertiary halides favor elimination, while primary halides favor substitution. (b)

$\text{CH}_3\text{Br} > \text{CH}_3\text{CH}_2\text{Br} > (\text{CH}_3)_2\text{CHBr} > (\text{CH}_3)_3\text{CBr}$ . Order strictly follows increasing steric hindrance which heavily blocks the incoming nucleophile. **Q45.** (i) Tertiary  $>$  Secondary  $>$  Primary ( $3^\circ > 2^\circ > 1^\circ$ ). (ii) Polar aprotic solvents (like acetone or DMF). (iii)  $\text{S}_\text{N}1$  mechanism. The leaving group departs to form a flat, planar carbocation intermediate. The nucleophile (

$OH^-$ ) can attack this planar intermediate from either the top face or bottom face with equal 50/50 probability, creating an equimolar racemic mixture of enantiomers. **Q46.** (a) Tert-butyl bromide forms a highly stable  $3^\circ$  carbocation (driving  $S_N1$ ), while its bulkiness completely blocks  $S_N2$ . Methyl bromide has zero steric hindrance (perfect for  $S_N2$ ) but forms a highly unstable primary carbocation (preventing  $S_N1$ ). (b) (i) Aniline

$\xrightarrow{NaNO_2 + HCl, 0^\circ C} C_6H_5N_2Cl \xrightarrow{Cu_2Cl_2/HCl} \text{Chlorobenzene}$ . (ii) Ethanol  $\xrightarrow{SOCl_2} C_2H_5Cl$ . Treat  $C_2H_5Cl$  with sodium acetylide ( $HC \equiv C^- Na^+$ )  $\rightarrow$  But-1-yne. (iii)

$2CH_3CH_2CH_2CH_2Cl + 2Na \xrightarrow{\text{Dry Ether}} \text{n-Octane}$ . **Q47.** (i) Saytzeff's Rule (Zaitsev's Rule). (ii) Pent-2-ene. (iii)

$CH_3CH_2CH_2CH(Br)CH_3 + KOH(alc) \xrightarrow{\Delta} CH_3CH_2CH=CHCH_3 + KBr + H_2O$

. Saytzeff's rule states that the highly substituted alkene is the major product because it is thermodynamically more stabilized by hyperconjugation (more alpha hydrogens). **Q48.** (i) Cannot break the strong H-bonds present in water. (ii) They stabilize the negative charge on the carbanion intermediate during nucleophilic attack via resonance. (iii) The C-Mg bond is highly polar, making the carbon heavily nucleophilic and basic. (iv) The nucleophile must attack exactly from the back-side of the leaving group to avoid electron repulsion, flipping the molecule like an umbrella. (v) Halogens withdraw electrons via -I effect (deactivating) but donate lone pairs via +R effect at ortho/para positions (directing). **Q49.** (a) (i)

$C_6H_5NH_2 \xrightarrow{NaNO_2/HCl, 0^\circ C} C_6H_5N_2Cl \xrightarrow{Cu_2Cl_2/HCl} C_6H_5Cl$ . (ii)

$C_6H_5Cl + CH_3Cl \xrightarrow{\text{Anhy. } AlCl_3} \text{o-chlorotoluene} + \text{p-chlorotoluene}$ . (b) A is tert-butyl chloride / 2-Chloro-2-methylpropane. B is 2,2,3,3-tetramethylbutane. Since A gives a completely symmetrical B with only one type of hydrogen, A must be highly branched. **Q50.** (i)

$CH_3CH_2Cl + POCl_3 + HCl$  (ii)  $X = C_6H_5N_2^+Cl^-$  (Benzene diazonium chloride);  $Y =$

$C_6H_5Cl$  (Chlorobenzene) (iii)  $CH_3CH_2CH_2Br$  (1-Bromopropane) (iv)

$CH_3CH_2NC + AgBr \downarrow$  (Ethyl isocyanide) (v)  $C_6H_5 - C_6H_5 + 2NaCl$  (Biphenyl)

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